

COMPLEX ENGINEERING PROBLEM
DESIGN OF MACHINES
ZIP LINE PULLY MECHANISM



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Abstract

This semester project focuses to design the zip line pulley mechanism in terms of applied shear force, engineering principles like Von Mises theory can be applied to ensure the pulley's material can withstand the stresses induced by the forces acting on it. Von Mises theory is often used in materials science and engineering to predict yielding of materials under combined loading conditions. The pulley system features integrated bearings specifically engineered to reduce friction during movement, thereby enhancing speed and reducing energy consumption. Additionally, the incorporation of specialized braking mechanisms, potentially utilizing magnetic or friction-based systems, allows for precise and rapid braking, ensuring a controlled and safe descent for users.

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CHAPTER 1: INTRODUCTION

1.1 Objectives

Performing stress analysis and Von Mises calculations for a zip line mechanism involves evaluating the structural integrity of the components under different loads and conditions. Here's a step-by-step guide to conduct this analysis:

- **Identify Components and Material Properties:**

List all the components of the zip line mechanism, including the pulley, frame, braking system, cables, etc. Determine the material properties for each component (e.g., tensile strength, yield strength, modulus of elasticity).

- **Define Loads and Constraints:**

Identify the loads acting on the system, such as the weight of users, dynamic forces during acceleration/deceleration, wind resistance, etc. Specify constraints or fixed points in the analysis, like attachment points to the supporting structures.

- **Create a 3D Model:**

Use software like SolidWorks, AutoCAD, or similar tools to create a detailed 3D model of the zip line mechanism.

- **Apply Loads and Boundary Conditions:**

Apply the identified loads and constraints to the 3D model. These could include forces, moments, and constraints on specific points or surfaces.

- **Stress Analysis:**

Run the analysis to calculate stress distributions within the components under applied loads. Identify areas of high stress concentration that may indicate potential failure points.

- **Von Mises Stress Calculation:**

Compute the Von Mises stress, which combines the normal and shear stresses to predict yielding in ductile material. Evaluate the Von Mises stress values across the components to identify critical areas prone to yielding or failure.

- **Validation Through the Calculations:**

In the calculation we run the simulation on the Solidwork of project then we do the mathematical calculation by hand then we compare the value. It almost near the error between both calculations is 5.2%

1.2 Methodology

1.2.1 Workflow

- **Problem Definition:**

We have to design the zip line pulley mechanism in terms of applied shear force, engineering principles like Von Mises theory can be applied to ensure the pulley's material can withstand the stresses induced by the forces acting on it. Von Mises theory is often used in materials science and engineering to predict yielding of materials under combined loading conditions. In the case of a zip line pulley, the forces acting on the pulley, such as the weight of the rider, friction, and dynamic forces during acceleration and deceleration, create shear stress on the pulley's material.

- **Literature Review:**

The zip line pulley mechanism is specifically designed for the smooth movement of riders along the cable in a zip line setup. **Functionality** The pulley is the central component that traverses the cable, allowing riders to move from a higher to a lower point using gravity. It's engineered for minimal friction and smooth gliding. **Design.** Typically made of durable materials like steel or aluminum, the pulley incorporates bearings or rollers to reduce friction and ensure efficient movement [1]along the cable.

- **System Identification:**

To analysis zip line pully mechanism in real life mechanism,[2] let us discuss the real-world example of a real-world example of a zip line pulley mechanism used in a recreational setting, such as a canopy tour or adventure park. Imagine an adventure park featuring a zip line course that spans across a forested area. Each zip line section comprises a series of towers or platforms anchored securely to trees or structural supports.

- **Load Determination:**

Identify and quantify the loads acting on the zip line mechanism User weight, dynamic forces during acceleration/deceleration, wind resistance, etc. Determine the varying loads under different operational conditions.

- **Constraints and Boundary Conditions:**

Define constraints and fixed points within the analysis Attachment points, areas of support or constraint [3]. Establish boundary conditions for realistic simulation.

- **Simulation Analysis:**

Employ simulation tools Use software like SolidWorks, to create a detailed 3D model of the zip line mechanism.

- **Practical Implementation:**

Conducting stress analysis and Von Mises calculations provides crucial insights into the structural integrity of the zip line mechanism [4]. Implementing this methodology ensures a safer and more reliable design by assessing and addressing potential failure points under various operating conditions.

1.3 Design Requirements

1.3.1 Material Properties:

We selected the material ASTM36 these are property of material as shown in diagram.

Property	Value	Units
Elastic Modulus	2e+11	N/m ²
Poisson's Ratio	0.26	N/A
Shear Modulus	7.93e+10	N/m ²
Mass Density	7850	kg/m ³
Tensile Strength	400000000	N/m ²
Compressive Strength		N/m ²
Yield Strength	250000000	N/m ²
Thermal Expansion Coefficient		/K
Thermal Conductivity		W/(m·K)
Specific Heat		J/(kg·K)

Figure 1 Material ASTM36 Property

1.3.2 Software Requirements:

Employ simulation tools Use software like SolidWorks, to create a detailed 3D model of the zip line mechanism. In Solidwork we see the stresses analysis and von misses. By using the tool and feature of the software.

CHAPTER 2: ANALYSIS ON STRESSES AND CALCULATION

2.1 Calculations:

2.1.1 Data:

$$S_{ut} = 4 \times 10^8 \text{ N/m}^2$$

$$S_{yt} = 2.5 \times 10^8 \text{ N/m}^2$$

$$F = 1000 \text{ N}$$

$$\text{Diameter of notch} = d = 2 \text{ cm}$$

$$\text{Width} = w = 80 \text{ cm}$$

2.1.2 Calculations:

$$\frac{d}{w} = 0.025$$

$$\frac{d}{h} = 4 \quad (\text{h denote the thickness})$$

Find σ_m

$$\sigma = \frac{My}{I} \quad (\text{where } I = \frac{1}{12} bh^3) \quad (1)$$

Now putting the value on the I, M and y then find the value of σ_m

$$I = \frac{1}{12} (w-d) \times h^3 \quad (2)$$

$$I = 8.125 \times 10^{-9} \text{ m}^4 \quad (3)$$

$$y_{\max} = \frac{h}{2} \quad (4)$$

$$y_{\max} = 0.0025 \text{ m} \quad (5)$$

$$M_m = \frac{F_m}{2} \times \frac{L}{2} \quad (6)$$

$$M_m = 56.25 \text{ Nm} \quad (7)$$

Now putting the value in the (1)

$$\sigma_{nom} = 17.30 \text{ MPa} \quad (8)$$

From graphs

$$K_t = 2$$

$$K_f = 1 + q(K_t - 1) \quad (9)$$

$$q = \frac{1}{1 + \frac{\sqrt{a}}{\sqrt{r}}}$$

$$q = 1.137 \quad (10)$$

Now putting the value in (9)

$$K_f = 2.137 \quad (11)$$

As we know the

$$\sigma = K_f \sigma_{nom}$$

$$\sigma = 3.6797 \text{ MPa} \quad (12)$$

Now according to the grid mesh independence table, we selected the

$$\sigma_{sim} = 3.9 \text{ MPa} \quad (13)$$

Now find the error by using the %error

$$\%Error = \frac{3.9 - 3.6}{3.9} \times 100$$

$$\%Error = 5.20\%$$

CHAPTER 3: SIMULATIONS ON SOLIDWORK AND RESULTS

3.1 Simulation of Stresses Analysis:

The simulations were run with upper body as fixed part of the zip line pulley mechanism while the load is attached to lower part where cable car is actually attached to the mechanism. After applying load, the thing that was noticed is maximum stress occurs at geometric discontinuities. Maximum stress occurs at notches. The maximum Von Misses stresses (*Figure 2 Simulation of Stresses Analysis*) are within the yielding strength so the material does not fail under the given loading conditions.

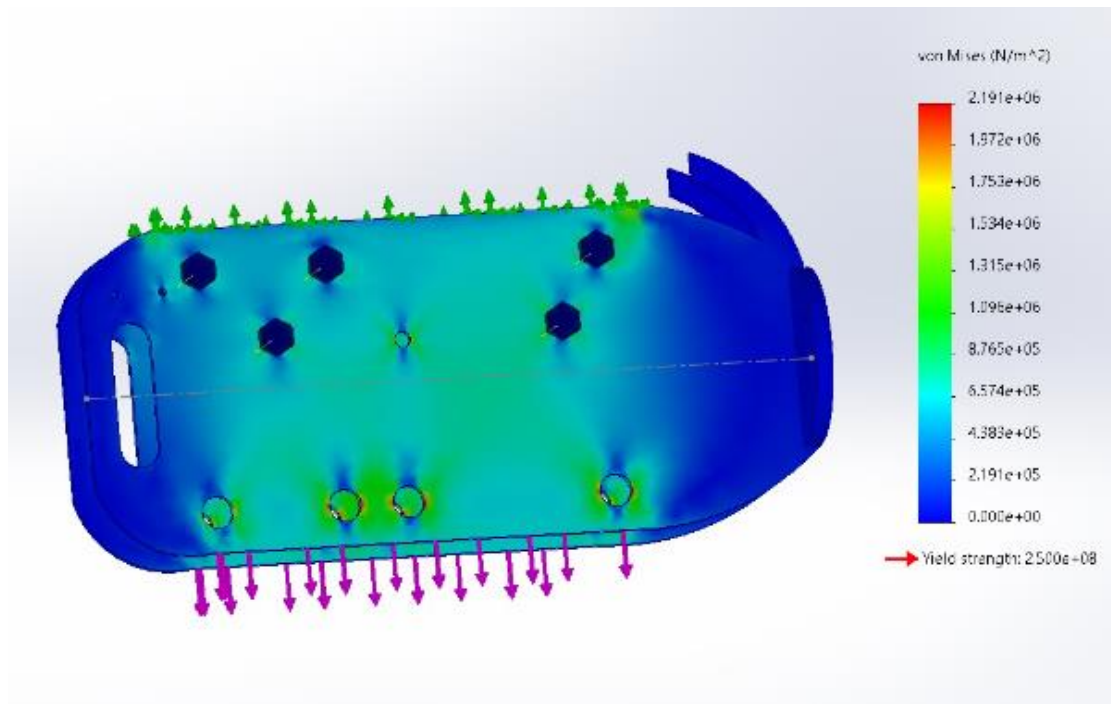


Figure 2 Simulation of Stresses Analysis

In displacement, the thing that was noticed was the maximum displacement occurs at the point where load is applied to the material. The other observation was that maximum displacement that occurs in the mechanism is 8.006×10^{-4} mm (*Figure 3 Displacement Analysis.*) This is very very small as compared to our mechanism. So, this is almost negligible. So, our mechanism is in safe zone.

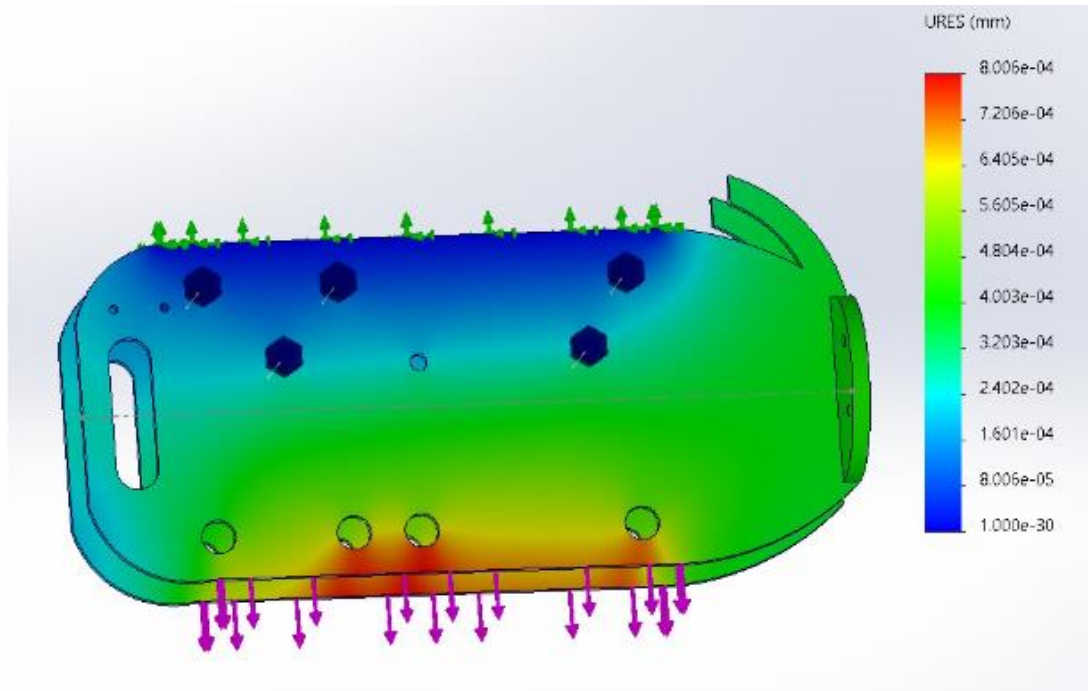


Figure 3 Displacement Analysis

3.2 Comparing Results of Simulation Applying von Mises Stresses with and Without Notches

The simulations of mechanism were run in two conditions. I.e with and without notches. According to those simulations (*Figure 4 Von Mises Stresses with Notches* *Figure 5 Von Mises Stresses without Notches*) it is observed that in case of notches, stress concentration occurs at notches and the material fails at notches. When simulations are run without notches, maximum stress occurs at the point with the geometric discontinuity. I.e at the top right corner of the mechanism.

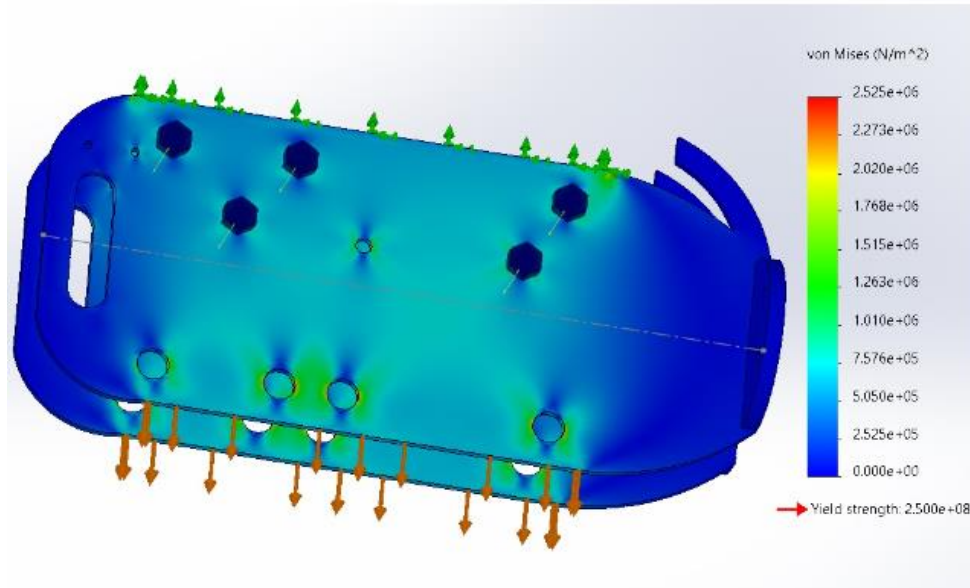


Figure 4 Von Mises Stresses with Notches

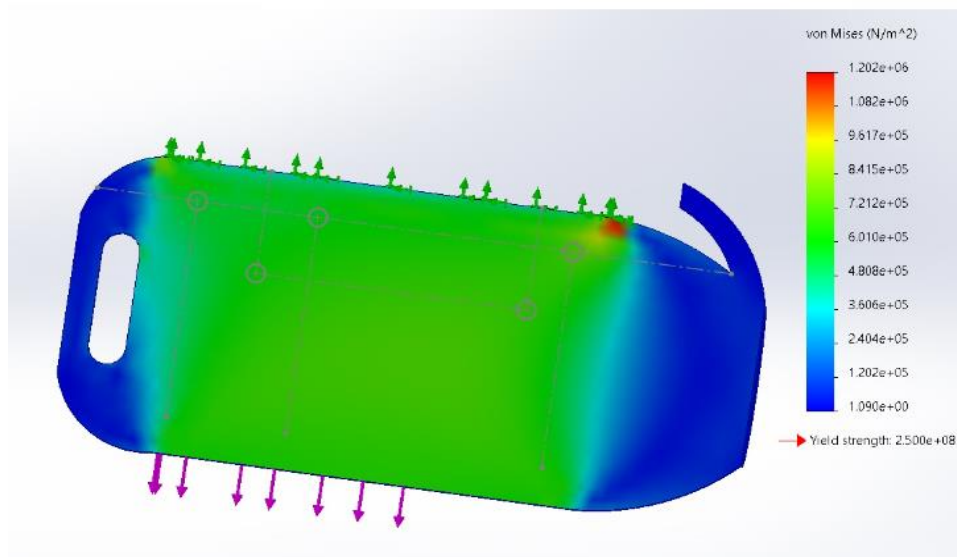


Figure 5 Von Mises Stresses without Notches

3.3 Grid Meshing Independence Mesh Size:

For grid independence, number of elements of mesh were increased from 35379 number of elements to 744947 elements (*Figure 6 Data Collection for Grid Independence*). The results were compared to check the relation. It was noted that results start to stabilize when our calculations reach 200000 Number of elements. So, readings at that point are considered to be values of our simulation results.

Study Number	Mesh Size (mm)	Number Of Elements	Maximum Stress (N/m ²)	Maximum Displacement (mm)
1	5	35379	2.22E+06	7.93E-04
2	4	43486	2.24E+06	7.93E-04
3	3	53971	2.20E+06	7.93E-04
4	2	81147	2.25E+06	7.95E-04
5	1	120325	2.28E+06	7.95E-04
6	0.5	186394	2.53E+06	7.99E-04
7	0.25	744947	3.90E+06	8.01E-04

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Figure 6 Data Collection for Grid Independence

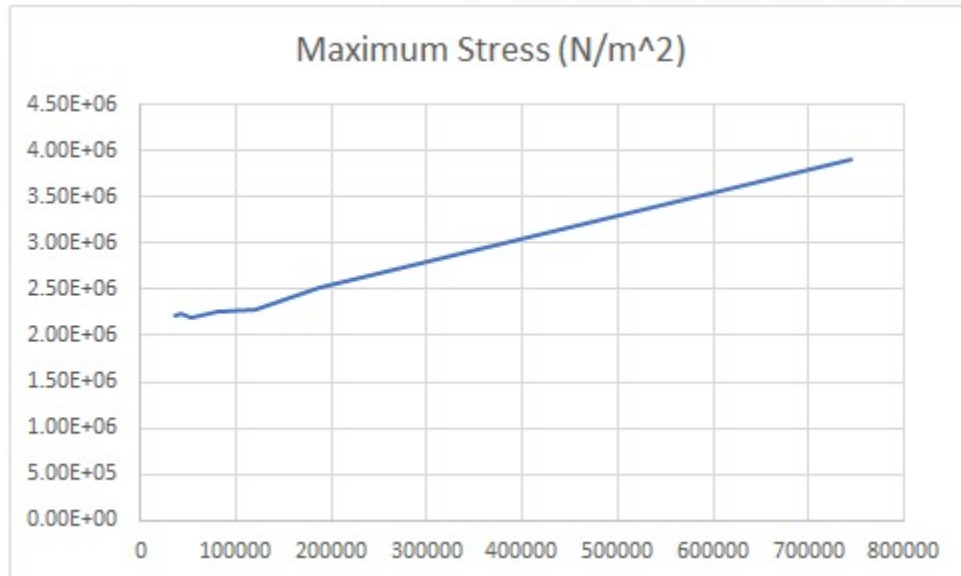


Figure 7 Graph of Maximum Stress (N/m²)



Figure 8 Graph of Maximum displacement (mm)

CHAPTER 4: HARDWARE

4.1 Hardware:

The hardware meets the software's system requirements [5] and specifications is crucial for conducting accurate and efficient stress analysis and Von Mises calculations on the zip line mechanism. Additionally, periodic updates and maintenance of both hardware and software are essential to maintain optimal performance.

4.1.1 Hardware Components:

- The operation of a pulley mechanism on a zip line is:
- The pulley mechanism is securely attached to the zip line cable, typically through a harness or a direct connection.
- Riders, equipped with necessary safety gear, such as harnesses or helmets, connect themselves to the pulley mechanism before starting the ride.
- When the rider releases the brake (if applicable) and begins their descent, gravity pulls them down the inclined cable.
- The pulley, designed with bearings or rollers, allows smooth gliding along the cable, minimizing friction and enabling the rider to move from the starting point to the endpoint.
- Braking mechanisms, controlled by the rider or an automated system, regulate speed and ensure a safe, controlled descent.



Figure 9 zipline hardware

4.1.2 Connect Hardware:

Connecting the necessary hardware for conducting stress analysis and Von Mises calculations for a zip line mechanism involves setting up a system that integrates the required computational resources and software tools which is the solid works.



Figure 10 Assembled Pulley Mechanism

4.2 Conclusion:

In CEA we learnt about the depends on various factors In conclusion, setting up a hardware system for stress analysis and Von Mises calculations for a zip line mechanism involves integrating computational resources, software tools, and ensuring proper connectivity to perform accurate simulations. By following a systematic approach, including hardware setup, software installation

4.3 References:

- [1] Y. Guo, Y. Yang, Z. Yang, and F. Ao, “Reliability Analysis of Zipline Project in A Mountainous Eco-tourism Scenic Spot,” in *IOP Conference Series: Earth and Environmental Science*, IOP Publishing Ltd, Mar. 2021. doi: 10.1088/1742-6596/1802/4/042073.
- [2] J. Vladić, T. Jojić, and R. Đokić, “Condition analysis and basis for selection of zipline arresting devices,” *IMK-14 - Istrazivanje i razvoj*, vol. 26, no. 4, pp. 89–94, 2020, doi: 10.5937/imk2004089v.
- [3] J. Vladic, T. Jojić, J. Vladić, and R. Đokić, “ANCHORAGE TYPE AND TENSION ROPE FORCE IMPACT ON ZIPLINE’S KINEMATIC CHARACTERISTICS,” *Trg Dositeja Obradovića*, vol. 11, no. 4, pp. 149–154, 2019, doi: 10.24867/MD.11.2019.4.149-154.
- [4] S. Vaucher, N. Yazdani, P. Felber, D. E. Lucani, and V. Schiavoni, “ZipLine: In-network compression at line speed,” in *CoNEXT 2020 - Proceedings of the 16th International Conference on Emerging Networking EXperiments and Technologies*, Association for Computing Machinery, Inc, Nov. 2020, pp. 399–405. doi: 10.1145/3386367.3431302.
- [5] T. Jojić, J. Vladić, and R. Đokić, “Zipline mathematical model forming, experimental verification and analysis of influential parameters,” *Archive of Applied Mechanics*, vol. 93, no. 11, pp. 4157–4180, Nov. 2023, doi: 10.1007/s00419-023-02487-0.